Olist Delivery Model

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Agenda

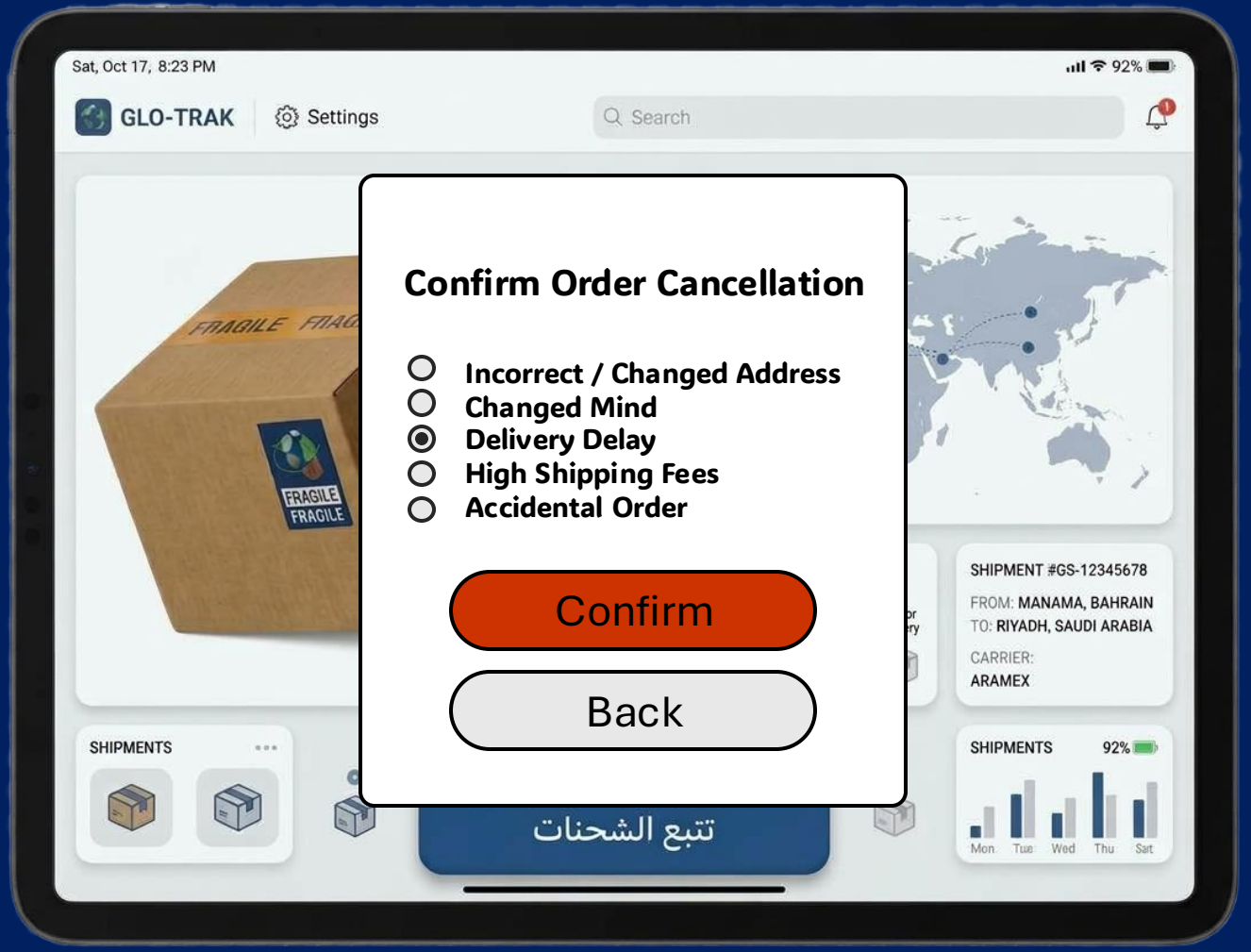
- **Business Section**
  - The raw issue
  - The structured objectives
- **Technical Section**
  - Correlation with the target
  - Data Processing Overview
  - Models MAE Comparison
- **Demo Section**
  - Live demo: the customer journey



The raw issue

Customers abandon carts when they cannot see when an order will arrive. Typical delivery takes 10 days, but 1 in 10 orders takes over 3 weeks.

Problem type: Regression





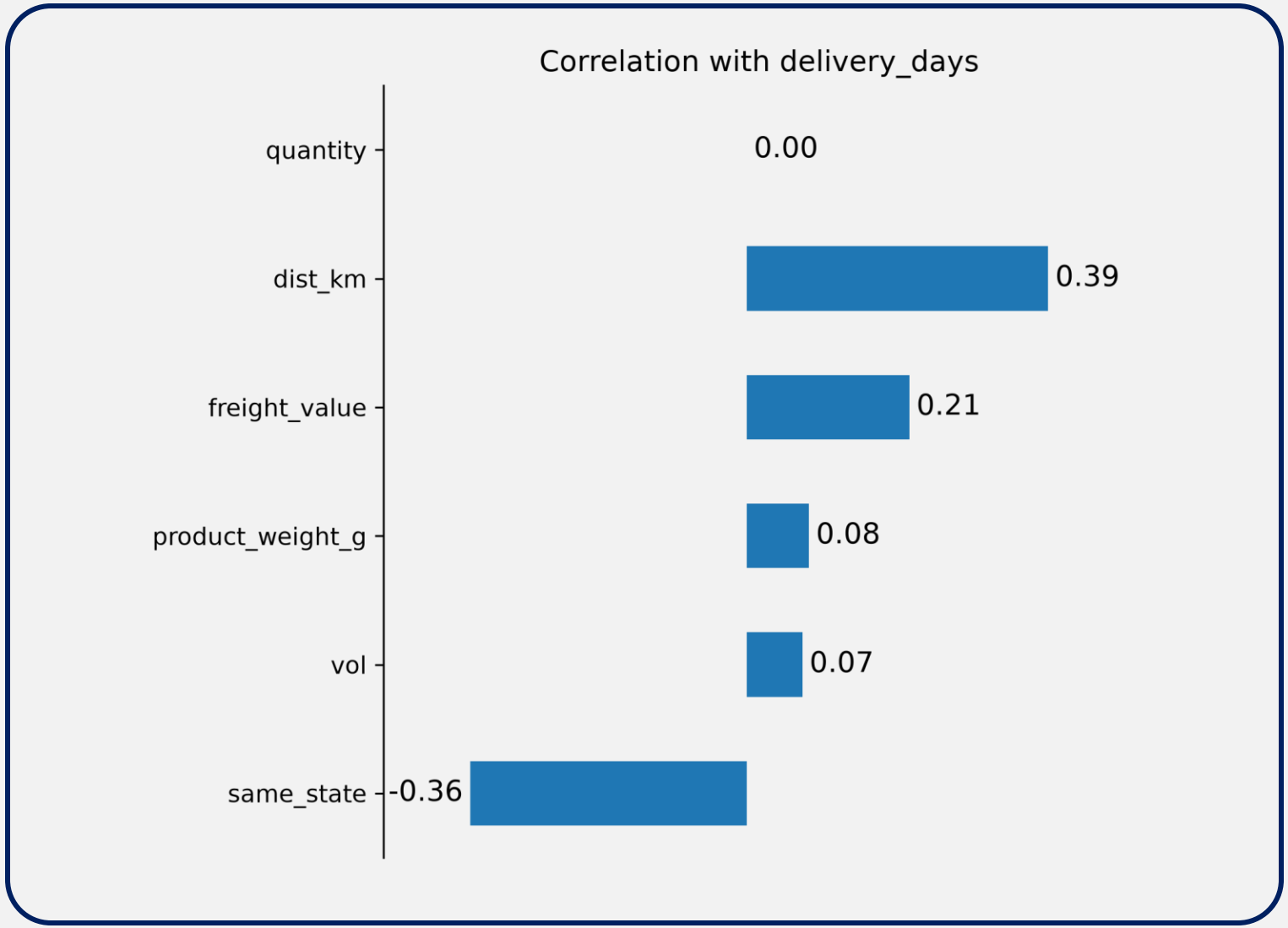
The structured objectives

- **Specific:** predict **delivery** time in days for every order at **checkout**, from **seller** and **customer location**, **distance**, product **size** and **weight**, **freight** cost and order **timing**.
- **Business objective:** show a trusted delivery **date before Order** Now, to cut order **cancellation** by **12% and increase sales** by gaining **customers satisfaction**.
- **Measurable: average** estimate within **5** days of the actual delivery (our model: **4.6 days** on unseen orders).

36%
Where customer and seller shares a state

0.39
Correlation between distance (km) and Delivery days

0.59 – 0.61
Correlation between the freight value and the product weight / volume



Data Processing Overview

Data

- **7 Olist files** combined into **one** table: one row per ordered item
- **Seller-to-customer distance** calculated from **zip** codes **averaged** coordinates
- Final dataset: **98,767** delivered orders

Cleaning

- Removed **contradictory** orders (“delivered” with no delivery date , “canceled” with a delivery date.)
- Removed items with **no product category**
- Merged repeat purchases of the same item into one row with a **quantity**
- Kept **delivered** orders only **98,797**, since we need a real delivery time to learn from
- We **replaced seller IDs** with simple labels (seller 1, seller 2, etc.)



Data Processing Overview



New features

Distance(km), same-state flag, product volume, purchase month, weekday and hour

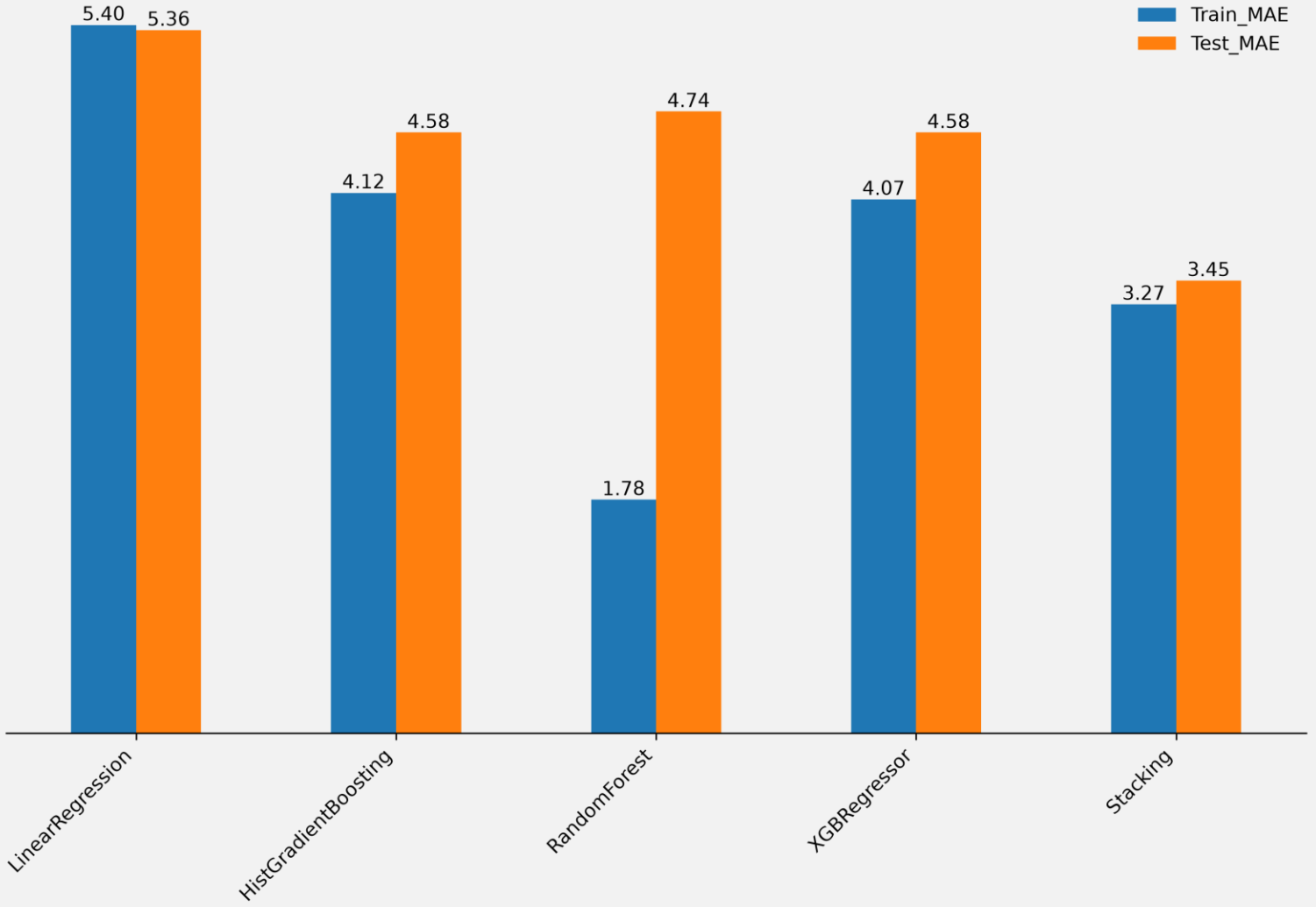
Missing information

- Rows missing **key** fields were **dropped**
- Smaller gaps (**weight**, **size**, **location**) were filled automatically, using the **'median'** value for numbers and "**missing**" for categories

Metric: Mean Absolute Error

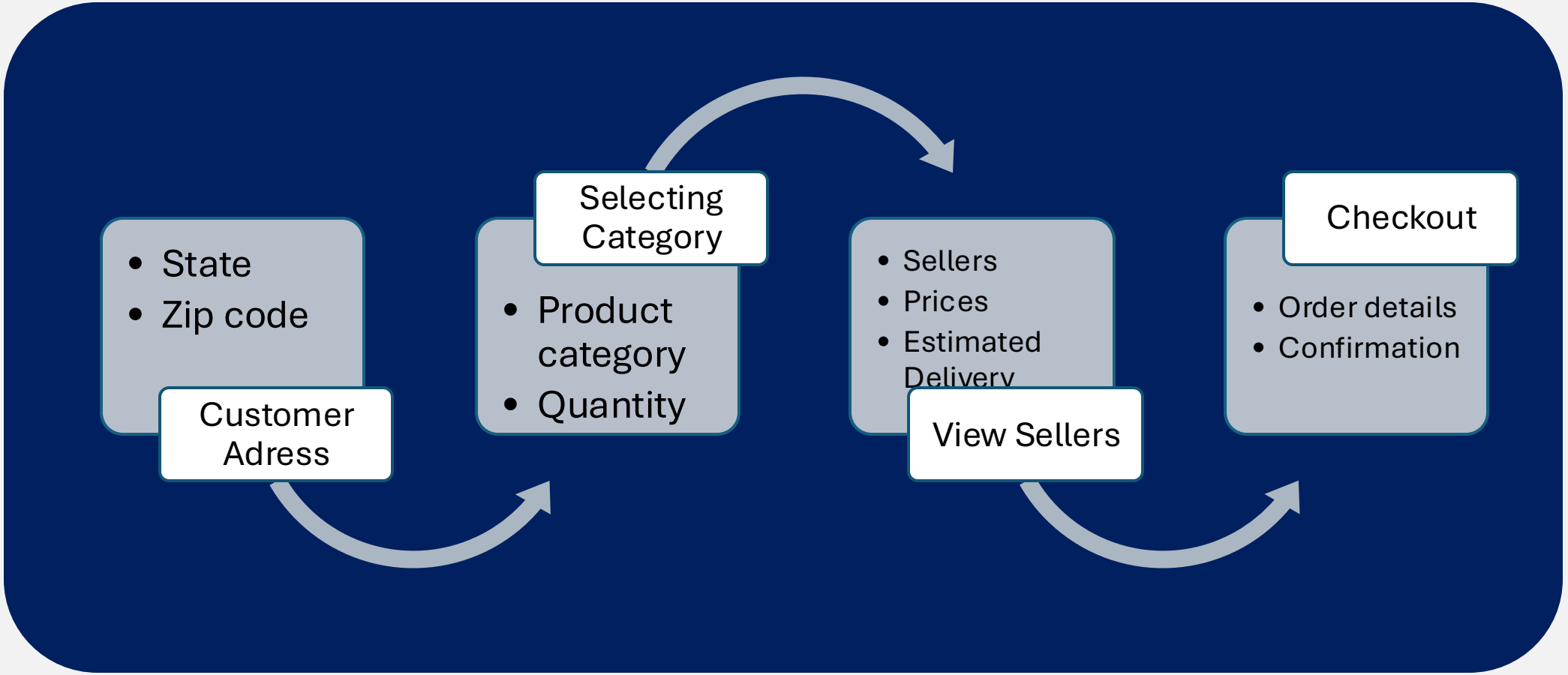
The **average** number of days our estimate is **off**. **Easy** to read, and it matches how customers **judge** a late or early delivery.

Models MAE Comparison





Live demo: the customer journey



Streamlit



Thank You